

1. (8 points) Use the mesh analysis method to solve the i in the circuit.

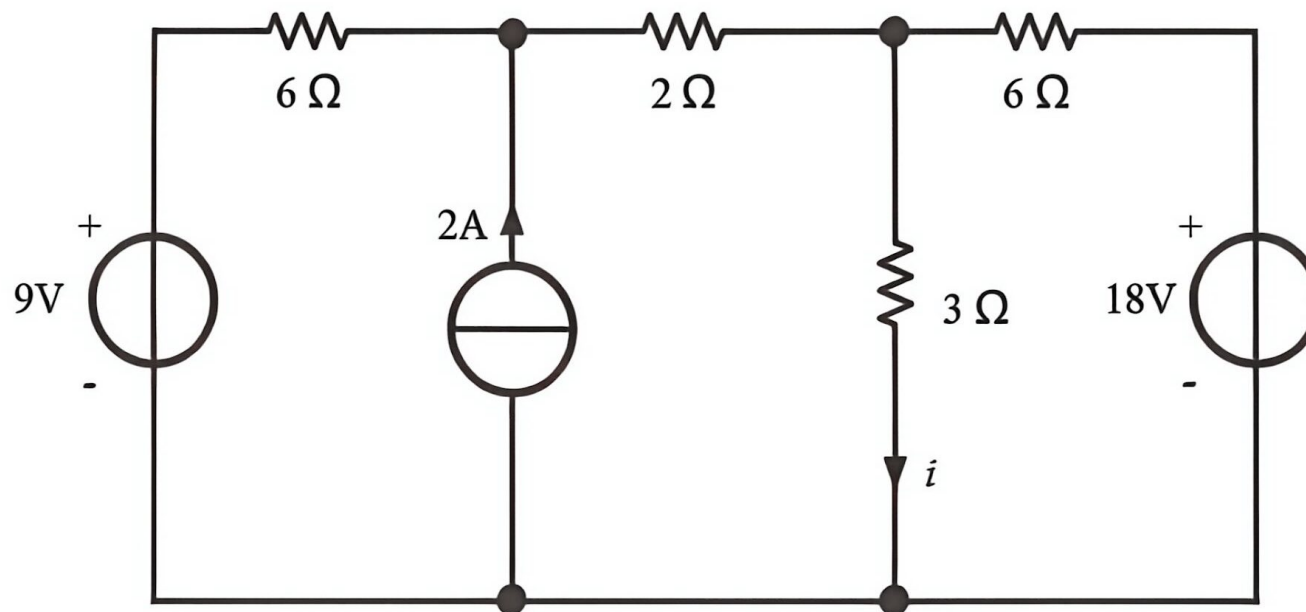


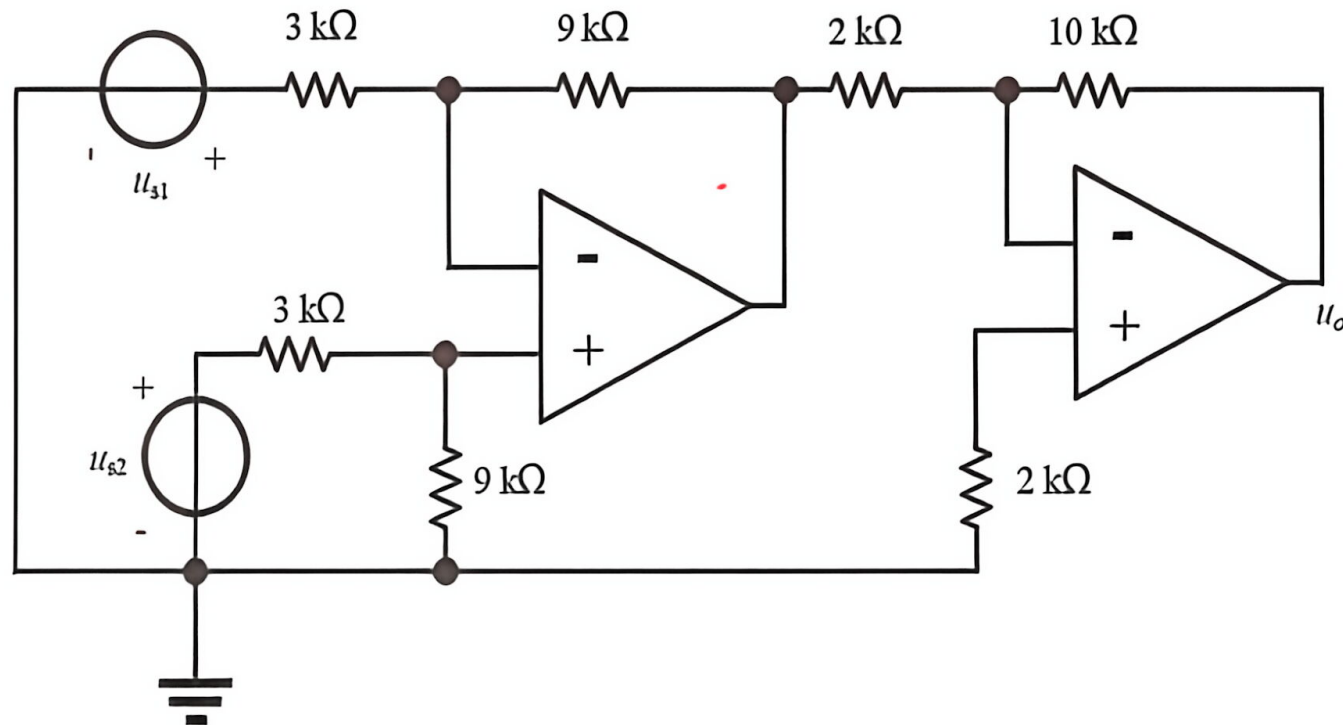
Figure 1



2. (10 points) The operational amplifiers in the circuit are ideal and operate in their linear range. Find

(1) the u_o if $u_{s1} = u_{s2} = 10 \text{ V}$; (5 points)

(2) the u_o if $u_{s1} = 10 \text{ V}$, $u_{s2} = 10.05 \text{ V}$. (5 points)



3. **(14 points)** The switch in the circuit has been turned off for a long time and the circuit has reached static state. The switch is turned on at $t=0$ s and then turned off at $t=3$ s. Find.

(1) the $u_c(t)$ for $t>0$; **(6 points)**.

(2) the power generated by the voltage sources for $t>0$ (note: write as functions of time). **(8 points)**.

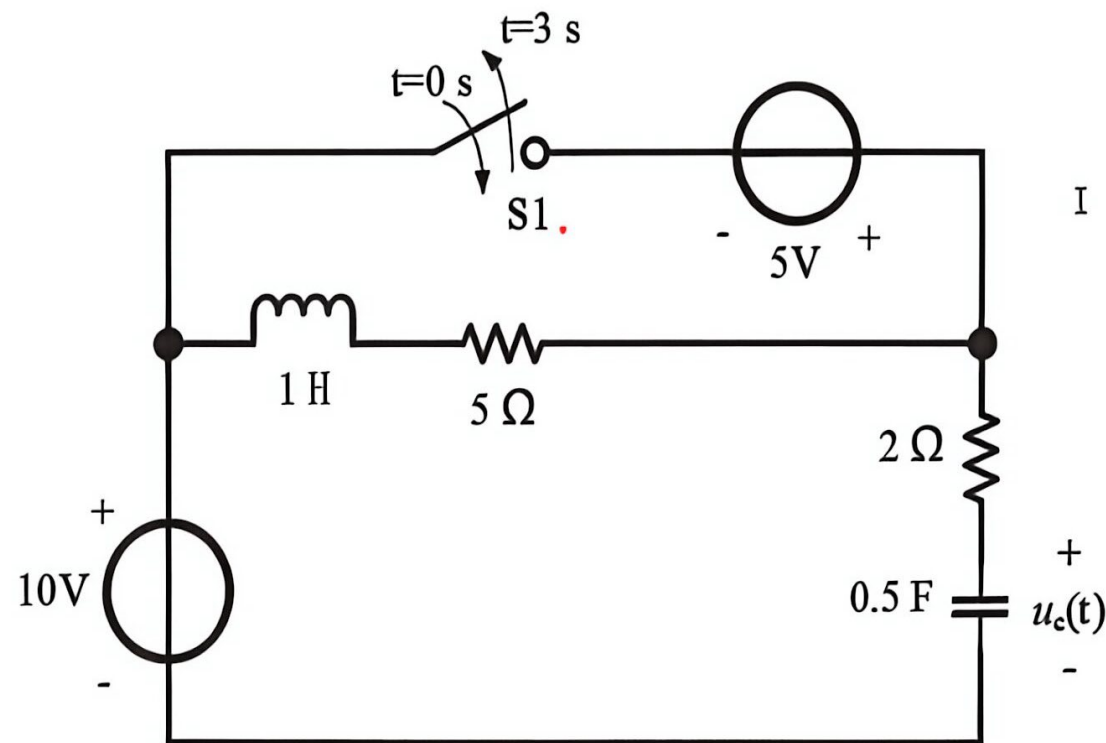
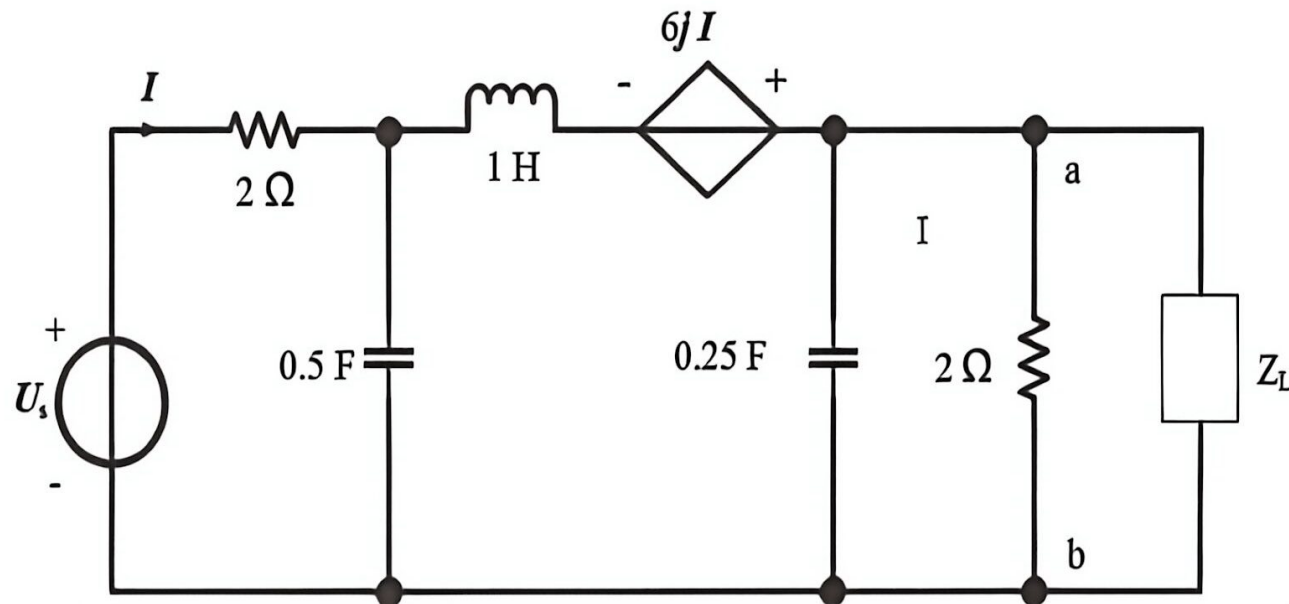


Figure 3.



4. **(14 points)** The circuit is in sinusoidal steady-state. The frequency, amplitude and initial phase angle of U_s are 1 rad/s, 5 V and 0° , respectively. Find.

- (1) the Thevenin equivalent at the terminals a-b of the circuit without the load impedance Z_L ; **(8 points)**.
- (2) the value of Z_L when it absorbs the maximum average power and the maximum average power absorbed by Z_L . **(6 points)**.

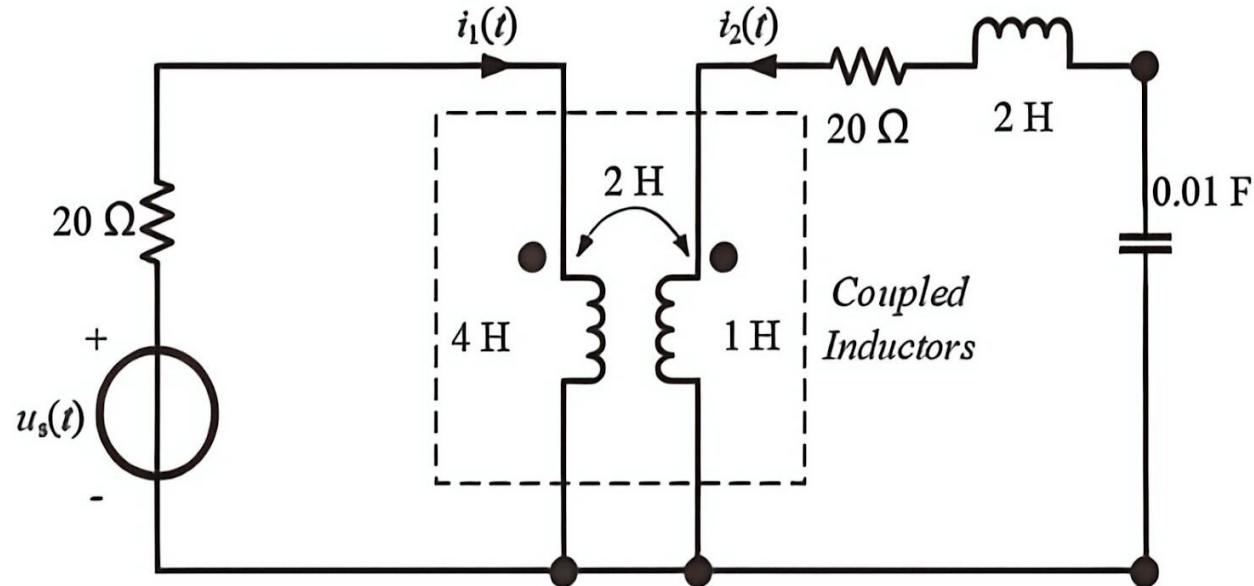


5. **(12 points)** The circuit shown in Figure 5 is in sinusoidal steady state. The voltage source $u_s(t) = 60\cos(10t + 45^\circ)$ V. The coupled inductors consist of 4H self-inductance, 1H self-inductance and 2H mutual inductance.

(1) Determine the coupling coefficient of the coupled inductors. **(2 points)**

(2) Find the current $i_1(t)$ and $i_2(t)$. **(8 points)**

(3) Find the total energy stored in the coupled inductors at time $t = 10\pi$ second. **(2 points)**



6. **(10 points)** The phasor domain three phase circuit is shown in Figure 6. The system is a balanced three phase system.

(1) Find the current phasors $\underline{I}_a$, $\underline{I}_b$ and $\underline{I}_c$. **(6 points)**

(2) Find the real power and the reactive power absorbed by the load of phase A (the impedance between node a and node n). **(4 points)**

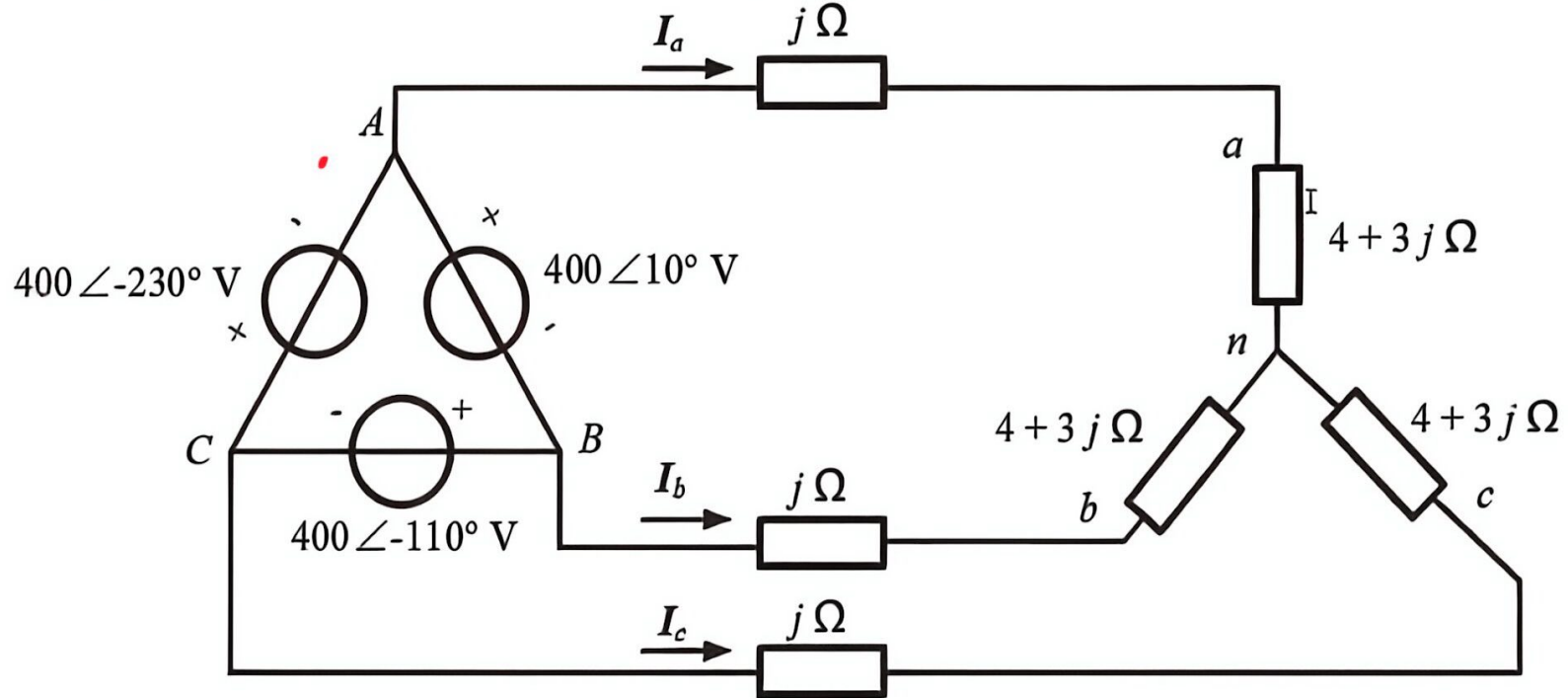


Figure 6



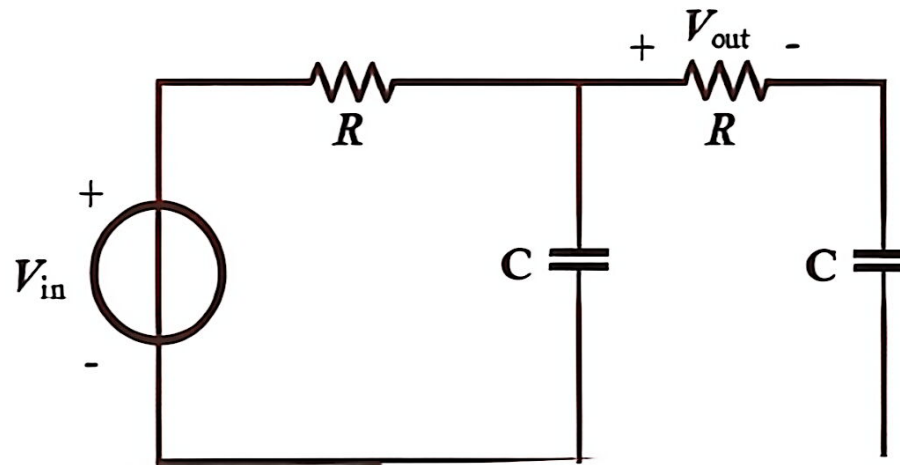
1. (14 points) The circuit is shown in Figure 7. The system input is V_{in} and the output is V_{out} . The transfer function $H(\omega) = \frac{V_{out}}{V_{in}} = \frac{j\omega/\omega_0}{(j\omega/\omega_0)^2 + k_1(j\omega/\omega_0) + k_2}$, where k_1 and k_2 are functions of the resistance R and the capacitance C .

(1) Express k_1 , k_2 and ω_0 as functions of R and C . (6 points)

(2) If $R = 10 \Omega$ and $C = 0.1 \text{ F}$, provide the Bode plots of $H(\omega)$ (including the magnitude plot and phase plot). Please label the corner frequencies, the slopes, the gains/phases of the plots. (6 points)

Hint: you can treat the expression $(j\omega/\omega_0)^2 + k_1(j\omega/\omega_0) + k_2$ as one term when depicting the Bode plot.

(3) Determine the filter type of $H(\omega)$. (2 points)



8. (18 points) The circuit is shown in Figure 8. At $t = 0^-$, $i_L(0^-) = 1 \text{ A}$, $v_C(0^-) = 0 \text{ V}$. The source $v_s(t) = 4e^{-3t} \text{ V}$.

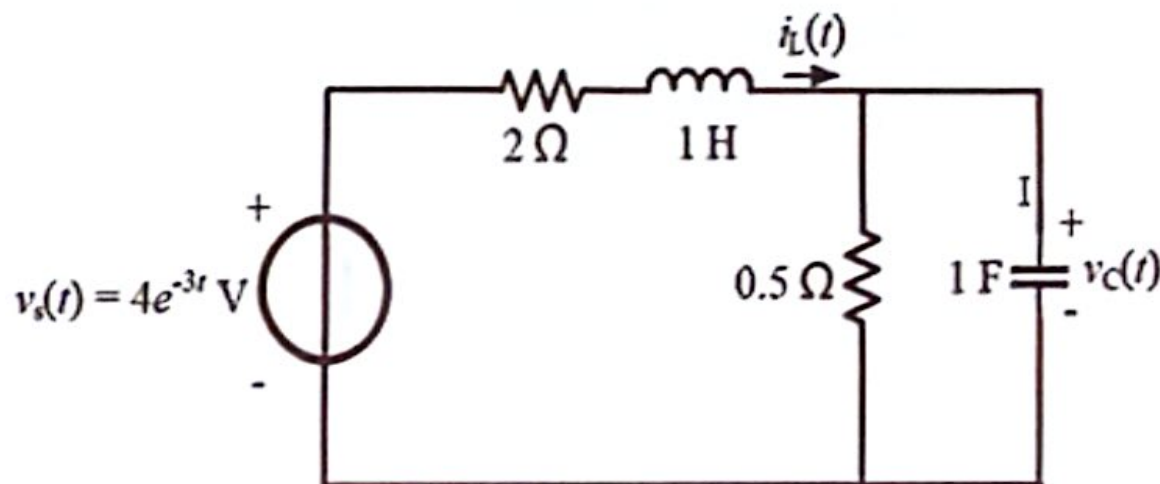


Figure 8.

- (1) Find $v_C(t)$ ($t > 0$) using time domain method. (8 points).
- (2) Find $v_C(t)$ ($t > 0$) using Laplace Transform method. (8 points).
- (3) What is the relationship between the results in (1) and (2)? (2 points).

